Academic Research Methodology (25.10.2026)
Department of Information Technology —
Instructor: Dalya Raad AlAni

LAB MANUAL 5: VISUAL ASSET DEPLOYMENT, NATIVE SUBCAPTION MANAGEMENT, & COMPLIANCE TEXT LAYOUTS

Lab Five

1 Part A: LaTeX Visual Infrastructure Assets Deployment

In computer science research and graduation theses, technical system artifacts—such as user authentication sequences, system block models, and network topologies—constitute primary empirical findings. Students must deploy these visual representations natively within the canvas layout without causing raster scaling defects or layout fragmentation.

1.1 The TikZ Native Vector Graphics Engine

To maintain publication-grade manuscript metrics, structural diagrams must be rendered as scale-invariant vector components instead of low-resolution raster objects (e.g., PNG or JPEG). Utilizing the `tikz` package framework layer enables programmatic definition of flowcharts, interfaces, and architecture layers directly inside the source code.

1.2 Subfigure Architecture and Labels Management

When technical investigations require side-by-side analytical comparisons—such as contrasting an isolated database layer with a modified cloud processing structure—the baseline model must employ the decoupled `subcaption` package environment. This setup avoids structural layout overflows and assigns nested tracking counters (e.g., (a) and (b)) completely automatically.

2 Part B: The Caption Tense Integration Workshop

Document navigation systems and reader focus depend heavily on clear caption rules. Figure descriptors must operate strictly within static parameter criteria.

2.1 Enforcing Static Descriptive Declarations

Unlike the experimental results section or the narrative implementation lifecycle steps, a figure caption must function as a standalone, timeless noun phrase. Students must actively scrub their captions of processing timelines:

- **Eliminating Active Past Frameworks**: Verbs describing past engineering actions (e.g., *showed, compiled, executed*) must be entirely omitted from structural descriptions.
- **Eliminating Speculative Future Elements**: Speculative future tracking strings (e.g., *will represent, will simulate*) break document compliance standards.
- **Verifying Static Integrity**: The caption area must represent an enduring state of structural truth via simple present tracking engines.

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3 Part C: Core Practical Case Studies

3.1 Case Study (a): Clean Multi-Tier Architecture Sandbox

Students will compile the following baseline code to verify that the coordinate system, package components, and shape rendering nodes handshake properly without overflowing the layout margins.

1. Launch TEXstudio and open a completely new empty editor canvas.
2. Copy, paste, and analyze the structured template script shown below:

```
\documentclass[12pt,a4paper]{article}
\usepackage[top=0.8in, bottom=0.8in, left=1.0in, right
   =1.0in]{geometry}
\usepackage{subcaption}
\usepackage{float}
\usepackage{tikz}
\usetikzlibrary{shapes.geometric, shapes.symbols,
   arrows.meta, positioning}

\begin{document}

        \section{System Methodology Infrastructure}
        The proposed data platform isolates user
           processing routines from tracking components
            using a secure access control boundary
           layer.

        \subsection{System Modeling Matrix}
        The physical distribution layer maps the
           structural properties across isolated
           environments as illustrated in Figure~\ref{
           fig:sys_mesh}.

        \begin{figure}[H]
                \centering
                \begin{subfigure}[b]{0.45\textwidth}
                        \centering
                        \begin{tikzpicture}[node
                           distance=1.2cm, auto, >=
                           Stealth, every node/.style={
                           transform shape, scale
                           =0.75}]
                                \node[draw, rectangle,
                                   fill=blue!10,
                                   rounded corners] (
                                   user) {User Client};
                                \node[draw, diamond,
                                   fill=orange!10,
                                   below=of user,
                                   aspect=2] (gate) {
                                   Auth Gateway};
                                \node[draw, cylinder,
                                   shape border rotate
                                   =90, fill=green!10,
                                   below=of gate,
                                   minimum width=1.4cm,
```

```
                                                minimum height=1.2
                                                cm, aspect=0.25] (db
                                                ) {Database Tier};

                                            \draw[->, thick] (user)
                                                -- node{\small
                                                Credentials} (gate);
                                            \draw[->, thick] (gate)
                                                -- node{\small
                                                Token Auth} (db);
                                    \end{tikzpicture}
                                    \caption{User authentication
                                        sequence.}
                                    \label{fig:auth_seq}
                                \end{subfigure}
                                \hfill
                                \begin{subfigure}[b]{0.45\textwidth}
                                    \centering
                                    \begin{tikzpicture}[node
                                        distance=1cm, auto, >=
                                        Stealth, every node/.style={
                                        transform shape, scale
                                        =0.75}]
                                            \node[draw, circle,
                                                fill=red!10] (edge)
                                                {Edge Node};
                                            \node[draw, rectangle,
                                                fill=gray!10, right=
                                                of edge] (proxy) {
                                                Proxy Layer};
                                            \node[draw, cloud, fill
                                                =cyan!10, cloud
                                                puffs=9, right=of
                                                proxy] (cloud) {
                                                Cloud Hub};

                                            \draw[<->, thick] (edge
                                                ) -- (proxy);
                                            \draw[<->, thick] (
                                                proxy) -- (cloud);
                                    \end{tikzpicture}
                                    \caption{Network topology
                                        layout.}
                                    \label{fig:net_layout}
                                \end{subfigure}
                                \caption{Proposed secure microservice
                                    environment: (a) automated GDPR-
                                    compliant token pipeline; (b)
                                    perimeter access routing
                                    infrastructure.}
                                \label{fig:sys_mesh}
                            \end{figure}

                        \end{document}
```

3. Save the source file as `lab5_baseline.tex` inside your course repository folder.

4. Press the **F5 key** on your keyboard to trigger the full Build & View pipeline execution.

3.2 Case Study (b): The Caption Tense Audit & Compliance Workshop

The Lab Task: Figure caption components must utilize purely static or present-tense frameworks. Review the extended architectural compliance configurations evaluated below:

- **Compliance Scenario 1 (GDPR Data Minimization Framework)**
  Incorrect Layout Caption: `\caption{The system layer cleared the GPS logs after variables were checked.}`
  Correct Compliance Layout: `\caption{Data minimization interface executing dynamic regex pipelines to sanitize system logs.}`
- **Compliance Scenario 2 (ISO/IEC 27001 Access Control Cryptographic Rules)**
  Incorrect Layout Caption: `\caption{UML schema which will map how cryptography will protect hardcoded server keys.}`
  Correct Compliance Layout: `\caption{Role-Based Access Control (RBAC) blueprint enforcing the principle of least privilege.}`
- **Compliance Scenario 3 (Academic Honesty & Code Attribution Mechanics)**
  Incorrect Layout Caption: `\caption{Figure 3.2 showed the open-source processing pipeline that we grabbed from GitHub.}`
  Correct Compliance Layout: `\caption{System architecture model utilizing external repository baseline dependencies via structured attribution parameters.}`

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Dalya Raad Al-Ani
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